SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning,*
and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

Duration: 1 Hour
QUESTIONS: 4 **Total: Marks:40**

Annexure A

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Code of Conduct:

I M. MUKKARAM_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
M. MUKKARAM

Annexure A

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

(a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.

(b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).

(c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

(a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.

(b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.

(c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

(a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.

(b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.

(b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.

(c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

AI/ML-Based Diabetes Diagnosis and Personalized Treatment Recommendation

Task 1: Data Preparation & Inductive Learning

(a) Inductive Learning Approach

Inductive learning learns general patterns from previously observed patient records and uses those patterns to predict the outcome for new patients.

For this problem, a **supervised classification approach** can be used. The model is trained using patient features and a known diabetes diagnosis.

Target variable:

- **Diabetes** – 0 = Non-diabetic, 1 = Diabetic.

Important features:

Feature	**Justification**
Glucose Level	High glucose is one of the strongest indicators of diabetes.
BMI	Higher BMI is associated with increased risk of type 2 diabetes.
Age	Diabetes risk generally increases with age.
Blood Pressure	High blood pressure is commonly associated with metabolic disorders and diabetes risk.
Cholesterol	Abnormal cholesterol levels can indicate metabolic risk.
Family History	Genetic/familial history can significantly influence diabetes risk.

The model learns relationships such as **high glucose + high BMI + older age → higher probability of diabetes**.

(b) Handling Class Imbalance

Suppose the dataset contains significantly more non-diabetic patients than diabetic patients. This creates **class imbalance**, which can cause the model to favor the majority class.

A suitable strategy is **SMOTE (Synthetic Minority Over-sampling Technique)**.

SMOTE creates new synthetic samples for the minority class rather than simply duplicating existing patients.

Advantages:

- Improves representation of diabetic cases.

- Helps the classifier learn minority-class patterns.
- Reduces the chance of ignoring diabetic patients.

SMOTE should be applied **only to the training data after the train-test split**, not before splitting, to prevent data leakage.

For a clinical application, **class weights** can also be used because missing a diabetic patient is more serious than incorrectly flagging a healthy patient.

(c) 70:30 Dataset Split and Preprocessing

The dataset can be divided into:

- **70% Training data** – used to learn the model.
- **30% Testing data** – used to evaluate its performance.

A **70:30 split** provides sufficient data for training while retaining a reasonably large independent test set. A **stratified split** should be used so that the diabetic/non-diabetic proportion is maintained in both sets.

Preprocessing

1. **Missing values:** Numerical missing values can be replaced using median imputation.
2. **Normalization:** Features such as glucose, BMI, cholesterol, and blood pressure can be standardized using:

$$z = \frac{x - \mu}{\sigma}$$

3. **Encoding:** Family history can be encoded as:
    - No = 0
    - Yes = 1
4. **Outlier checking:** Extremely abnormal measurements should be investigated.
5. **Class balancing:** SMOTE can be applied to the training set.

Normalization is especially useful for **Logistic Regression and other distance/gradient-based models**, while Decision Trees generally do not require feature scaling.

Task 2: Decision Tree for Diagnosis

(a) Decision Tree Classifier

A **Decision Tree** can classify patients as diabetic or non-diabetic by repeatedly dividing patients according to their features.

The root node is selected using **Information Gain** or **Gini Index**.

Gini Index

$$Gini = 1 - \sum_{i=1}^{n} p_i^2$$

A lower Gini value represents a purer node.

Illustrative first three levels

An example tree could be:

```
              Glucose Level
             /           \
         ≤ 126            > 126
          /                 \
        BMI               Diabetes
       /   \
    ≤ 25     > 25
     /          \
 Non-Diabetic   Age
               /  \
           ≤ 45     > 45
            /         \
    Non-Diabetic   Diabetic
```

The actual tree must be generated from the supplied dataset.

Example root-node comparison

Feature	**Illustrative Gini after split**
Glucose	**0.32**
BMI	0.39
Age	0.42
Blood Pressure	0.46
Cholesterol	0.48

Since **Glucose produces the lowest Gini value**, it would be selected as the root in this example.

Alternatively, using Information Gain, the feature with the **highest Information Gain** would be selected.

(b) Model Evaluation

The model should be evaluated using:

Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision

$$Precision = \frac{TP}{TP + FP}$$

Recall

$$Recall = \frac{TP}{TP + FN}$$

F1-Score

$$F1 = 2\frac{Precision \times Recall}{Precision + Recall}$$

Illustrative results

Metric	Decision Tree
Accuracy	84%
Precision	78%
Recall	**89%**
F1-Score	83%

These values are examples and **not actual results from the patient's dataset**.

False Negatives

A **False Negative (FN)** occurs when:

Actual patient = Diabetic
Model prediction = Non-diabetic

False negatives are particularly important in diabetes diagnosis because a missed diabetic patient may not receive timely testing, monitoring, or treatment.

To reduce false negatives:

- Increase the importance/weight of diabetic cases.
- Use SMOTE on training data.
- Adjust the classification threshold.
- Optimize for **Recall/Sensitivity** rather than accuracy alone.
- Perform additional clinical testing for high-risk patients.

In healthcare, **high recall is often more important than maximizing accuracy**, because missing a genuine diabetes case can have serious consequences.

Task 2(c): Post-Pruning

A fully grown Decision Tree can **overfit** the training data. Pruning removes unnecessary branches.

Pre-pruning

Pre-pruning restricts tree growth using parameters such as:

- Maximum depth
- Minimum samples per leaf
- Minimum samples for splitting

Post-pruning

Cost-complexity pruning balances tree complexity and classification error:

$$R_\alpha(T) = R(T) + \alpha \mid T \mid$$

where:

- $R(T)$= error of the tree
- $\mid T \mid$= number of terminal nodes
- α= complexity parameter

Illustrative comparison

Model	Accuracy	F1-Score	Tree Complexity
Pre-pruned Tree	82%	80%	Medium
Post-pruned Tree	**85%**	**84%**	Low

Post-pruning is generally preferable for clinical deployment because the resulting tree is **simpler, easier to interpret, and less likely to overfit**.

However, the final choice should be based on validation performance, especially **recall, sensitivity, AUC, and calibration**, rather than accuracy alone.

Task 3: Statistical Learning for Risk Stratification

(a) Logistic Regression

Logistic Regression is suitable because diabetes prediction is a binary classification problem.

The model estimates:

$$P(Diabetes = 1) = \frac{1}{1 + e^{-z}}$$

Where,

$$z = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n$$

The coefficients indicate how each feature influences diabetes risk.

Illustrative comparison

Metric	Decision Tree	Logistic Regression
Accuracy	84%	**87%**
AUC-ROC	0.86	**0.91**

These are **illustrative values**.

When is a statistical model preferable?

Logistic Regression is preferable when:

- The relationship between predictors and outcome is reasonably stable.
- Probability/risk estimates are important.
- Interpretability is required.
- Doctors need to understand how individual features influence risk.
- The dataset is not extremely complex.

For example, the coefficient of glucose can indicate whether increasing glucose is associated with increased diabetes probability.

(b) Feature Importance Analysis

Decision Tree – Top 3

An illustrative ranking may be:

1. **Glucose Level**
2. **BMI**
3. **Age**

Logistic Regression – Top 3

Based on standardized coefficient magnitude:

1. **Glucose Level**
2. **BMI**
3. **Family History**

Comparison

Both models agree that **Glucose and BMI** are major predictors. The third feature may differ because Decision Trees capture **non-linear threshold relationships**, while Logistic Regression assumes a more systematic relationship between predictors and the log-odds of diabetes.

Therefore, disagreement does not necessarily indicate that one model is wrong.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) MDP Formulation

The treatment recommendation system can be modeled as a **Markov Decision Process (MDP)**.

1. States

States represent the patient's health condition.

Example:

State	Description
S1	Low-risk / healthy
S2	Moderate-risk
S3	High-risk

State	Description
S4	Diabetic – controlled
S5	Diabetic – poorly controlled

The state can be determined from glucose, BMI, blood pressure, symptoms, previous treatment response, and other clinically relevant information.

2. Actions

The available actions are:

- **A1 – Diet**
- **A2 – Exercise**
- **A3 – Medication**
- **A4 – Monitor**

3. Reward Function

The reward represents improvement in patient health.

Example:

$$R = Health_{t+1} - Health_t$$

Illustrative reward structure:

Outcome	Reward
Significant health improvement	+10
Moderate improvement	+5
No significant change	0
Health deterioration	-10
Serious adverse outcome	-20

In a real medical system, rewards must be designed with clinicians and should strongly penalize unsafe outcomes.

4. Transition Dynamics

The transition describes how a patient's condition changes after an action.

For example:

Moderate Risk + Exercise → Low Risk

Moderate Risk + Diet → Low/Moderate Risk

High Risk + Monitor → High Risk

High Risk + Medication → Controlled Diabetes

The transition probabilities should ideally be learned from longitudinal clinical data rather than invented.

Task 4(b): Q-Learning

Q-Learning learns the expected value of taking an action in a particular state.

The update equation is:

$$Q(s,a) \leftarrow Q(s,a) + \alpha\left[r + \gamma \max_{a'} Q(s',a') - Q(s,a)\right]$$

Where:

- α= learning rate
- γ= discount factor
- r= reward
- s'= next state

Assume:

$$\alpha = 0.5,\ \gamma = 0.9$$

Initially all Q-values are zero.

Iteration 1

Suppose:

State S2 + Exercise → S1, reward = +5.

$$Q(S,2Exercise) = 0 + 0.5[5 + 0.9(0) - 0]$$
$$\boxed{Q(S,2Exercise) = 2.5}$$

Another transition:

S3 + Medication → S4, reward = +10.

$$Q(S,3Medication) = 0.5[10 + 0.9(0)]$$
$$\boxed{Q(S,3Medication) = 5}$$

Iteration 2

Suppose the best Q-value in S1 is now 4.

For:

S2 + Exercise → S1, reward = +5:

$$Q(S,2Exercise) = 2.5 + 0.5[5 + 0.9(4) - 2.5]$$
$$= 2.5 + 0.5(6.1)$$
$$\boxed{Q(S,2Exercise) = 5.55}$$

Suppose:

S3 + Medication → S4, reward = +10 and the maximum Q-value of S4 is 6.

$$Q(S,3Medication) = 5 + 0.5[10 + 0.9(6) - 5]$$
$$\boxed{Q(S,3Medication) = 10.2}$$

Example Q-table

State	Diet	Exercise	Medication	Monitor
S1 Low Risk	2	4	0	3
S2 Moderate	3	**5.55**	1	2
S3 High Risk	2	3	**10.2**	4
S4 Controlled	3	5	2	**6**
S5 Poorly Controlled	1	2	**8**	0

These values are illustrative.

Five patient episodes

The agent can be trained for at least five episodes, for example:

Episode 1: S2 → Exercise → S1

Episode 2: S3 → Medication → S4

Episode 3: S2 → Diet → S1

Episode 4: S5 → Medication → S4

Episode 5: S4 → Exercise → S1

Convergence Criterion

Training can be considered converged when:

$$\max_{s,a} | Q_{new}(s,a) - Q_{old}(s,a) | < \epsilon$$

for a predefined small value such as:

$$\epsilon = 0.01$$

over several consecutive episodes.

Task 4(c): Final Learned Policy

The optimal policy is obtained by selecting the action with the highest Q-value:

$$\pi(s) = \arg\max_{a} Q(s,a)$$

An illustrative final policy is:

Patient State	Recommended Action
S1 – Low Risk	Monitor
S2 – Moderate Risk	Exercise
S3 – High Risk	Medication*
S4 – Controlled Diabetes	Exercise
S5 – Poorly Controlled	Medication*

*In an actual clinical system, medication decisions **must not be autonomously prescribed by an RL agent**. They require clinician oversight and consideration of contraindications, current medications, laboratory results, patient preferences, and clinical guidelines.

Ethical Considerations

1. Patient Safety

An incorrect recommendation can cause serious harm. The RL agent should therefore operate under **strict clinical safety constraints** and clinician supervision.

2. Bias

If the training data underrepresents certain age groups, genders, ethnic groups, or socioeconomic populations, the model may produce biased recommendations.

3. Explainability

Doctors should be able to understand **why a recommendation was generated**, especially when medication is involved.

4. Privacy

Patient information is sensitive. Data should be anonymized, securely stored, and processed according to applicable healthcare privacy regulations.

5. Human-in-the-Loop

The AI should function as a **clinical decision-support system**, not as an independent doctor. A qualified healthcare professional should have the final authority.

6. Continuous Monitoring

The deployed system should be regularly evaluated for changes in accuracy, bias, safety, and patient outcomes.